

Master Manual: Smart Water Pump Controller System

1. Executive Summary

The **Smart Water Pump Controller** is an industrial-grade IoT solution designed for deep well water management in Leon, Iloilo. By integrating high-power electrical components (Magnetic Contactors, Thermal Overloads) with modern microelectronics (ESP32, Ultrasonic Sensors), the system provides automated tank management, real-time remote monitoring, and mission-critical hardware protection.

2. System Architecture

The project is divided into three distinct layers, ensuring that a failure in the software layer does not compromise the physical safety of the motor.

I. The Muscle Layer (Hardware)

- **Protection:** 20A MCB and LR2-D13 Thermal Overload Relay.
- **Switching:** CJX2-2510 Magnetic Contactor.
- **Power:** Handles the 220V AC high-current path for the 1.5HP Lotus pump.

II. The Brain Layer (Firmware)

- **Logic:** ESP32 (38-pin) running a non-blocking state machine.
- **Sensing:** JSN-SR04T (Level) and YF-G1 (Flow).
- **Control:** 5V Opto-isolated relay module.

III. The Soul Layer (Cloud/Software)

- **Data:** Firebase Real-Time Database (RTDB).
- **Interface:** Web-based dashboard for manual override and status visualization.

3. Mission-Critical Logic

The system follows a "Safety-First" priority hierarchy:

1. **Level 1 (Physical):** If the TOR senses heat > 104 , it physically breaks the coil circuit. This is the ultimate failsafe.
2. **Level 2 (Software):** If the ESP32 detects 0 flow for $> 30s$ while the pump is active, it initiates a **Dry-Run Lockout**.
3. **Level 3 (Manual):** The external bypass switch allows for human intervention regardless of the ESP32's state.

4. Installation & Site Standards

- **Environment:** IP65 Waterproof Enclosure.

- **Signal Integrity:** $1k\Omega/2k\Omega$ voltage dividers must be installed at the box termination point to protect the 3.3V GPIO pins from 5V sensor logic.
- **Transmission:** 10m-20m UTP Category 6 cable for sensor data, utilizing twisted pairs for power/ground to minimize voltage drop.

5. Maintenance Schedule

- **Monthly:** Verify Ultrasonic sensor accuracy against physical dipstick measurement.
- **Quarterly:** Inspect PG16/PG9 cable glands for insect ingress or seal degradation.
- **Bi-Annually:** Perform a "Tug Test" on high-current terminals to ensure thermal cycling hasn't loosened connections.